The Summary Never Reads the Mask: Localisation-Grounded Report Faithfulness in a Deployed Chest Radiograph System

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Abstract

Deployed chest radiograph systems increasingly show a diagnostic prediction, a localisation overlay, and a short natural-language summary together, and the overlay is read as evidence that the summary is grounded in the image. We audit one such deployed four-class system (Normal, pneumonia, tuberculosis, COVID-19; DenseNet-121 classifier, Grad-CAM localisation compressed into a U-Net for one-forward-pass serving). Its summary generator turns out to be a four-entry static template keyed only on the predicted class: no property of the mask, not the side, the region, the extent, nor even whether a region exists, reaches the printed text. Over the system's 245 already-scored test overlays we then show, with no retraining, that gating the summary on measured mask properties changes what can honestly be said; that on most films the honest output is to say nothing about location, because the overlay sits outside the lung fields; and that permuting each film's mask onto a different film drops claim accuracy to near chance (15%). That last result is the failure mode the current template's silence on location happens to sidestep and the one a naive descriptor-driven generator would walk straight into. We report this on a deployed system and give a small evaluation protocol, crossing heatmap source with generator type, for the general question of when a generated radiology report is reading the image it describes.

1 Introduction

Chest radiograph systems built for low-resource triage bundle three outputs behind one click: a diagnostic label, a heatmap that claims to show where the evidence lies, and a short clinical-language summary. Each is usually evaluated on its own, if at all. Classification accuracy is scored against a held-out set, localisation quality against some overlap reference, and the sentence is treated as presentation. That split hides a specific, checkable failure: a system can be accurate, can show a heatmap that looks like an explanation, and can still print a summary whose spatial claims, when it makes any, land on the mediastinum or the cardiac silhouette instead of the lung.

We audit a system with exactly that shape. LungLens is a four-class chest X-ray classifier on a DenseNet-121 backbone [11], with Grad-CAM [21] localisation compressed into a U-Net [20] so an overlay costs one forward pass at serving time, not a forward-plus-backward pass. Every prediction carries a summary. Reading the source settles what writes it: a four-entry lookup keyed on the predicted class, joined to one of five fixed sentences chosen by whether the prediction is confident, uncertain, or Normal. No coordinate, area, side, or component count from the mask is ever an input. Nothing else feeds it, so the question of whether the localisation subsystem contributes anything to the report has a direct answer, and Section 4 is built to give it.

We turn that question into a measurement. Reusing the region-level statistics the system's own failure analysis already computes over 245 scored test films (side, in-lung fraction, area fraction, plus a component count we add), we build a small descriptor-to-prose pipeline and compare it with the deployed template under four heatmap conditions (no mask, the served Grad-CAM map, the served U-Net map, and a mask lifted from a different film) crossed with two generators: the static template and one gated on the descriptors. The in-lung gate suppresses a spatial claim on 62% of films, because the served overlay's centre of mass falls outside the lung fields on a majority of them. Permuting the mask onto a different film leaves the claim rate unchanged but drops the claim's accuracy against the film's own ground truth to 15%, the sanity check the saliency literature uses to tell a method that reads the input from one that does not [1].

Contributions. (1) An audit, from source, of what a deployed chest-radiograph summary generator conditions on, and a direct answer to whether its localisation subsystem reaches the text. (2) A retraining-free descriptor pipeline over the system's own overlays with an in-lung gate, and a measurement of how often that gate fires on real served overlays. (3) A degenerate-mask sanity check in the tradition of Adebayo et al. [1], applied to the generated report rather than the saliency map alone. (4) A small evaluation protocol crossing heatmap source with generator type, cells we could fill without new training or clinician time filled and the rest left open.

2 Related work

Classification and its explanations. CNN interpretation of chest radiographs matured with CheXNet [19], a 121-layer DenseNet at radiologist level for pneumonia on ChestX-ray14 [27]; DenseNet-121 [11] stays competitive with residual [10] and compound-scaled [24] backbones at chest-radiograph data scales, which is why it is a fixed component here, not a contribution. Four-class Normal/pneumonia/TB/COVID work exists [7, 14], as do multi-task designs sharing a representation across classification and segmentation [9]. Grad-CAM [21] and its variants [5] are the default explanation for such models [23], and a body of work asks whether the maps are faithful to the model or only plausible: Adebayo et al. [1] show several saliency methods survive randomising the model's weights, and Kindermans et al. [15] extend the concern. Our degenerate-mask control carries that test from the map to the report built on it: if an unrelated map leaves the text's claims as accurate, the map is not doing the work the text implies.

Generated reports. A separate line generates full narrative reports from radiographs [6, 13], usually scored with text-overlap metrics that reward fluency without checking image support. Factual-consistency metrics over clinical entities [17], region-grounded generation that ties each sentence to an image region [25], and biomedical vision-language pretraining [3] all respond to that gap. LungLens predates this: its summary is closer to a caption, and we ask the same faithfulness question at the point a much simpler system first gets the chance to fail it. Earlier hand-crafted CXR features [4, 8, 12] and the COVID-era CNN surveys [2, 16, 18, 22, 26] are background for the backbone and dataset choices below.

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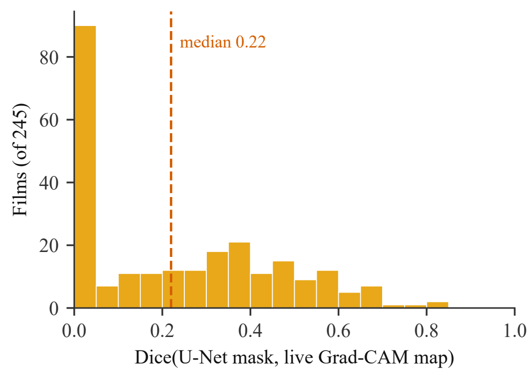

Figure 1: Dice between the served U-Net mask and a freshly computed live Grad-CAM map, same film and class, over all 245 served films. The distribution is heavy at zero, not merely low on average: a generator that reads a location from one of these two sources cannot assume the other agrees with it.

3 System

Data. The classifier trains on eight public chest-radiograph sources, deduplicated by content hash to 37,553 images and sampled to 32,000, arranged so each of the four classes draws from at least two acquisition pipelines, a deliberate hedge against the source-fingerprinting failure of Zech et al. [28]; Section 6 reports the residual. Images are border-cropped 8% at train and inference time to drop burned-in corner text, resized to 224×224, and ImageNet-normalised. The split is patient-grouped and reproducible: because every directory listing is sorted before use, re-running split construction from the raw files reproduces the test accuracy to four significant figures on the same 4,759 held-out images. That guarantee is not free. An earlier, unsorted version of the same code moved the reported test accuracy by nearly two points purely as a function of filesystem directory-traversal order.

Classifier. A DenseNet-121 [11], ImageNet-initialised, fine-tuned under class-weighted cross-entropy (inverse-frequency weights, mean 1.0) for up to 40 epochs with early stopping on validation macro-F1, Adam at 10^{-4}, batch 32. It is a fixed upstream component; its full evaluation is in Appendix A.

3.1 Localisation: explanation amortisation, not knowledge distillation

A second network, a U-Net-style encoder-decoder (`MultiTaskUNet`: four `DoubleConv` blocks, $64 \to 128 \to 256 \to 512$ channels, skip connections, 7,705,224 parameters, counted from the served checkpoint), learns to reproduce the classifier's Grad-CAM map in one forward pass, avoiding the backward pass Grad-CAM needs. The frozen DenseNet (no gradient enters it here) produces a Grad-CAM map at `denseblock4.denselayer16.conv2` for each image's *ground-truth* label, ReLU'd, resized to 224×224, and divided by its own maximum; targets are held in memory for the run and never cached to disk. The U-Net is trained against that target with an unweighted Dice+BCE loss (Adam, 3×10^{-4}, batch 8) over 3,000 images for 20 epochs; the served channel is the class being visualised.

We call this *explanation amortisation*, not knowledge distillation. Nothing is shared between the two networks: no layers, no logit or feature matching, no gradient path from student to teacher. The only channel is one 224×224 per-class scalar field, computed once and discarded. Very little survives a channel that narrow, and the cost is measurable: on the 245 served films, the median Dice between the U-Net's mask and a fresh live Grad-CAM map for the same film and class is 0.22, with 27% showing no overlap and 15% above 0.5 (Figure 1). What the compression buys is a real serving-cost saving (Section 5); stating that saving together with this cost is the honest description.

3.2 Report generation

The version under review disposed of this subsystem in eleven words. Section 5's central finding needs its exact behaviour, so we specify it here.

What it is. A static Python dictionary keyed on the predicted class (Normal, Pneumonia, Tuberculosis, Covid-19), joined to one of five fixed sentences selected by three booleans: is the prediction below the confidence threshold, does the visualised class equal a confident non-Normal prediction, and was a mask computed at all. No model, prompt, decoding parameters, or learned weights; the full listing is Appendix B.

Inputs. The argmax of the softmax output and its probability; a flag for whether the visualised class is the top prediction; a flag for whether that top prediction is Normal; and, for the trailing provenance line only, which subsystem produced the mask. No coordinate, area, side, zone, or component count from the mask is ever read.

Confidence gate. Below 60% top-class probability the report reads "Uncertain" with a fixed cautionary sentence; the only other non-standard path is a caught exception on malformed input.

Failure this produces. The "confident non-Normal" branch does not check whether a region survived the display threshold, so the text can assert "the outline marks the most influential area" on a film where nothing was outlined. This happens on 2 of the 245 scored films, one a confidently wrong tuberculosis call (95.7%) on a radiograph whose true label is Normal, with no thresholded region anywhere in the image (Appendix C).

4 Experimental design

Classification accuracy and localisation quality are already measured (Section 3, Appendix A). The new experiment asks one thing: does the summary change, and change correctly, as a function of the heatmap it is given. We cross a heatmap condition with a generator condition over the 245 test films already scored for the failure analysis (all 102 pneumonia-predicted-Normal errors, all 92 Shenzhen and 12 Montgomery test films, and 40 stratified correct films).

Heatmap conditions. H0, no mask (class and confidence only). H1, the live Grad-CAM map. H3, the served U-Net map. H5, a degenerate control: the mask of a uniformly random *different* film under a fixed derangement (seed 42), so each film keeps its own class and confidence but gets someone else's spatial evidence.

Generator conditions. S0, the deployed generator (Section 3.2), rebuilt from source. S1, a template stating side, vertical third, extent, and focality from the region descriptors, gated to make no claim when the thresholded region's in-lung fraction is below 0.5 or when no region survives the display threshold. S2, a text-only LLM given class, confidence, and descriptors, is out of scope here (Section 6).

Region descriptors. For each film the served overlay's thresholded region (the app's own thresholds and cleaning: 0.50 for a U-Net mask, 0.45 for Grad-CAM, 5×5 elliptical opening, components under 0.3% of the frame dropped) is reduced to laterality (left/right/outside both, from the centroid against a separate lung-field segmenter), vertical zone (upper/mid/lower third, from the centroid; new here), extent (focal $<5\%$, patchy 5–25%, extensive $>25\%$, from area fraction), focality (single vs. multifocal, from component count; new here), and in-lung fraction (the gate). Laterality, extent, and in-lung fraction were already in the system's analysis code; the two new descriptors were checked against it by matching recomputed area fractions to a median absolute difference of 0.0001 over the 204 films with a region.

Cells run (no GPU, no training): S0 as baseline; S1 crossed with H0, H_actual (whichever of H1/H3 the system served for that film), and H5. A per-film H1-versus-H3 pairing, computing both maps fresh for every film, needs the 245 radiographs staged for a forward-plus-backward pass each and is

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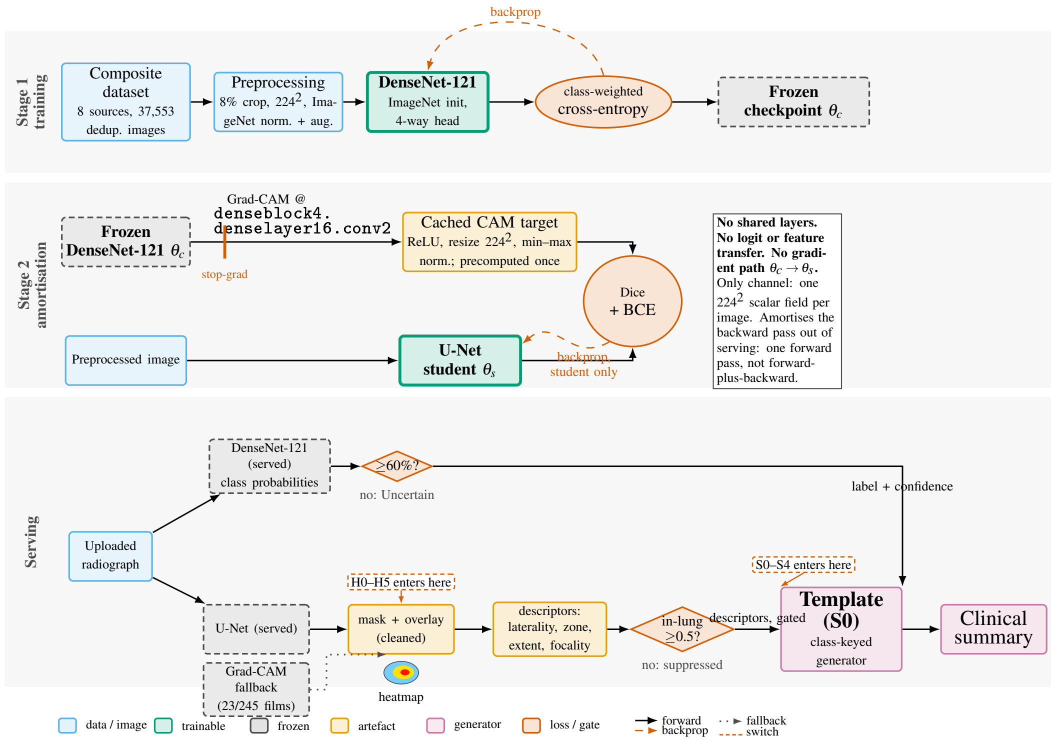

Figure 2: System overview. Stage 1 trains the served DenseNet-121 classifier. Stage 2 supervises a U-Net student on Grad-CAM targets cached from the frozen classifier (stop-gradient marked): no layers, logits, or features are shared, so the only channel is one 224×224 scalar field per image, amortising the backward pass out of serving. At inference the classifier supplies label and confidence; the localiser supplies gated region descriptors; both arrive at the report generator, a class-keyed template (S0), which alone determines the printed summary. Tags mark where H0–H5 and S0–S4 enter.

left for a revision.

Metrics. Spatial-claim rate (how often a claim is made). Grounding precision (does a claim match the film's descriptors). Phantom-claim rate (does the generator assert evidence that did not survive the threshold). Anatomical plausibility (does a claimed location fall in the lung, true by construction under the gate). H5 grounding accuracy (does a claim from another film's descriptors match this film's truth). Latency, reported in prose.

5 Results

Classifier, in one paragraph. The DenseNet-121 reaches 96.18% accuracy and 95.77% macro-F1 on the patient-grouped 4,759-image test set, with per-class recall from 92.2% (pneumonia) to 98.5% (Normal). The dominant error, 102 pneumonia films predicted Normal (56% of all 182 errors), sits in a single source folder whose label, *lung opacity*, is a broader radiological finding than pneumonia, and several of those films have clear lung fields on inspection: label noise as much as model failure. Full detail is in Appendix A.

5.1 The summary does not move with the mask, because it never reads it

Under every heatmap condition (H0, H1, H3, H5), S0 prints byte-identical text for a given film, because none of its five branches take mask geometry. The explainability apparatus and the report

Table 1: The evaluation grid. Filled cells report a measured number (Section 5); open cells are proposed and not run. S0 makes no spatial claim under any heatmap, so its row collapses by construction, which is the headline. S1's heatmap cells are reported jointly as H_actual; a per-film H1-vs-H3 split is left for a revision.

Generator (rows) / Heatmap (cols)	H0 (none)	H1 (Grad-CAM)	H3 (U-Net)	H5 (permuted)
S0 (deployed template)	measured	measured	measured	n/a (static)
S1 (descriptor template)	measured	measured	measured	measured
S2 (text LLM)	open	open	open	open
H2 (Grad-CAM++), H4 (second backbone), S3/S4 (VLM): open, see Section 6				

S0 (deployed). Prediction: Covid-19 (99.1%). Findings consistent with Covid-19. Warmer areas indicate possible bilateral ground-glass opacities. Heatmap: region driving the Covid-19 prediction. The outline marks the most influential area. *Overlay source: disease-segmentation U-Net.*

S1 (descriptor, H_actual). Prediction: Covid-19 (99.1%). Findings consistent with Covid-19. Warmer areas indicate possible bilateral ground-glass opacities. Heatmap evidence: left lung, mid zone, patchy, multifocal. *Overlay source: disease-segmentation U-Net.*

(Reconstructed from the template in Appendix B and the film's measured descriptors; `figures/paired_summary.pdf` overrides.)

Figure 3: The same radiograph, predicted Covid-19 at 99.1% confidence, under both generators. S0 is identical whether or not a mask is available; S1 adds a spatial claim built from the film's own descriptors. Both are language over measured pixels, not a clinical finding.

apparatus are disconnected. In Table 1 this is the collapsed S0 row: a property of the generator, not of our harness.

5.2 Gating on measured evidence changes what can honestly be said

S1, gated on in-lung fraction, states a spatial claim on 92 of 245 films (37.6%). On the other 153, either no region survives the display threshold (41 films, 16.7%) or the region's centre of mass is outside the lung fields. The gate rate follows directly from an already-published measurement: 55% of the served overlays put the majority of their region outside the lung (Appendix D). Figure 3 shows a pair on one film: S0's silence on location above, S1's grounded "left lung, mid zone, patchy, multifocal" below, read from the film's own descriptors. Without localisation the generator can only restate the class; with it, it can also say where, and the accuracy of that "where" is bounded by Section 3.1's Dice.

5.3 The degenerate-mask control

Permuting each film's mask onto a different film under a fixed derangement leaves S1's claim rate unchanged (92/245, since a permutation preserves the marginal of which films pass the gate), but the claim's accuracy against the film's true side, zone, and extent falls to 14/92 (15.2%), near the chance level the skewed zone and side marginals imply. This is Adebayo et al. [1]'s check moved from the saliency map to the report a naive descriptor-driven generator would write: if the stated location is no more accurate with the right mask than the wrong one, the localisation subsystem is not grounding the sentence. The same bound acts on any real generator drawing from H1 or H3: the two maps disagree at median Dice 0.22 (Section 3.1), so a generator reading one of them inherits that ceiling. A localisation-grounded report is exactly as trustworthy as the localisation under it, and no more.

Latency. On a CPU laptop (i7-13620H, 10 threads), the served pipeline averages 293 ms (sd 81) when the distilled U-Net supplies the mask, the common case at 91% of test films, and 560 ms (sd

22) on the live Grad-CAM fallback (forward-plus-backward, 9%), a test-set-weighted 318 ms. The roughly 2× gap is what the U-Net buys on hardware with no GPU. Report generation is a dictionary lookup and adds nothing measurable; a hosted-API generator would need its network latency reported separately.

6 Proposed protocol and limitations

What is proposed, not measured. Table 1's open cells are not implied results. H2 (Grad-CAM++) and S3/S4 (a vision-language model given the radiograph, or the radiograph plus the overlay) are inference- or API-cost work scoped out by design. H4, Grad-CAM from a second, independently trained backbone, is the most informative open cell, since it separates "the summary tracks the pathology" from "the summary tracks this model," and it needs a second classifier trained end to end. No clinician was available; the automated metrics are faithfulness proxies, not substitutes. An LLM-as-judge score, if used later, must report its agreement with human labels on a subsample or not be used.

Dataset-source bias. Despite multi-source classes and identical 8% cropping, per-source test accuracy runs from 83.3% (Montgomery, 12 images) to 99.5% (TBX11K). The two weakest sources fail oppositely in the localisation data: Shenzhen's overlays are diffuse (median 41% of the frame on TB-positive films), Montgomery's are mostly absent (a quarter of its films produce no region). Source and class stay partly confounded in an eight-source composite, so 96.18% is an optimistic upper bound, not a clean prospective number.

Label noise. 82 of the 102 pneumonia-to-Normal errors come from a folder whose label, *lung opacity*, is a mapping decision rather than a diagnosis, and several inspected films have clear lung fields. Some of this error mode is model failure and some is the model correctly reading a film that should not have been labelled positive; the two are not separable after the fact without the original read.

Reproducibility. An earlier, unsorted directory-traversal order moved the split from 22,419/4,785/4,796 to 22,420/4,759/4,821 images and a re-scored checkpoint from 96.62% to 98.11%, nearly two points from enumeration order alone on an otherwise identical pipeline and seed. Sorting every listing before use is now load-bearing for every number here being reproducible.

No clinical validation. LungLens is a research and educational system, not evaluated prospectively against radiologist reads. Its summary, static template or descriptor-gated, is language over measured pixels, not a clinical finding, and the deployed interface should say so, not only this paper.

7 Broader impact

A confident label, an apparently explanatory heatmap, and a fluent summary shown together are more persuasive than their parts justify, especially to a non-specialist in the low-resource setting this system targets. Our central finding, that a plausible summary can be entirely disconnected from the localisation evidence beside it, is a caution against that persuasiveness, not an endorsement of shipping the descriptor-gated version as a clinical tool. Any future report layer here should carry the non-diagnostic disclaimer this paper argues is missing, and be checked for faithfulness before fluency.

8 Conclusion

LungLens pairs an accurate classifier, a fast localisation shortcut, and a clinical-language summary. Auditing the third from source shows it never reads the first two: the summary depends only on the predicted class and its confidence. A small retraining-free gate on the system's own region statistics shows what changes when it does read them, and most of the time the honest output is to say nothing about location, because the evidence sits outside the lung. The degenerate-mask control makes the

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point: without a faithfulness check, a more sophisticated report generator would not be safer, only more convincing.

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A Classification detail

The classifier is a fixed upstream component (Section 3); this is its full evaluation.

Table 2: Per-class performance of the served DenseNet-121 on the held-out test set ($n = 4{,}759$; accuracy 96.18%, macro-F1 95.77%). Percentages except support. Derived from the confusion matrix below.

Class	Prec.	Recall	F1	Spec.	Supp.
Normal	95.30	98.50	96.90	94.60	2,502
Pneumonia	98.30	92.20	95.10	99.30	1,478
Tuberculosis	94.70	95.10	94.90	99.70	243
Covid-19	95.70	96.60	96.20	99.50	536
Macro avg.	96.00	95.60	95.77	98.30	4,759

Table 3: Confusion matrix on the held-out test set, from `chest_classifier_metrics.json` for the served checkpoint. The repository `README.md` carries a different matrix that does not match this file on cells other than the headline pneumonia-to-Normal count; the matrix here is the one behind every classification number in this paper.

True (rows) / Pred. (cols)	Normal	Pneumonia	Tuberculosis	Covid-19
Normal	2,465	14	9	14
Pneumonia	102	1,363	4	9
Tuberculosis	10	2	231	0
Covid-19	10	8	0	518

The dominant error is one-directional: 102 pneumonia radiographs predicted Normal, 56% of all 182 misclassifications, and no other off-diagonal cell exceeds 14. 82 of the 102 come from the COVID-19 Radiography Database's `Lung_Opacity` folder, which `app.py` maps to Pneumonia though "lung opacity" is broader than pneumonia; 8 come from its `Viral Pneumonia` folder and 12 from the paediatric pneumonia set. A left–right lung-field mean-intensity probe supports reading this as label noise: median absolute difference 5.9 grey levels on these 102 films, against 8.9 on correctly classified pneumonia and 4.0 on true Normals, and only 22% of the 102 exceed the 90th percentile of the true-Normal distribution, against 37% of correctly classified pneumonia, so the model's outputs track the pixels rather than miss something present.

Source-bias probe. An earlier composite drew three of four classes from one source each, so a model could score by recognising the scanner rather than the pathology [28]. The composite here weakens that: tuberculosis, Covid-19, and pneumonia each draw from at least two sources, though the COVID-19 Radiography Database still supplies three classes through one pipeline. Per-source accuracy spans 83.33% (Montgomery, 12 images) to 99.50% (TBX11K), wider than prevalence alone explains, though the two extremes are also the two smallest partitions, so sampling noise and genuine source sensitivity cannot be separated here. The 96.18% headline is read as inflated by acquisition cues to an unknown degree.

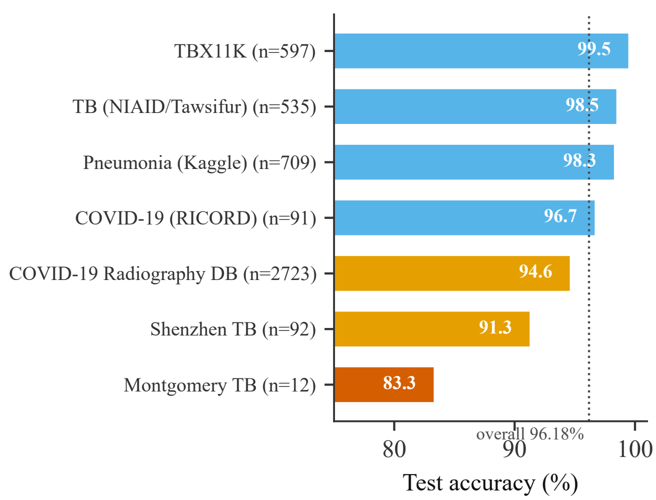

Figure 4: Test accuracy by source dataset, sorted, with support (n) per source and the overall figure marked. The two weakest sources are also the two smallest partitions (92 and 12 images), so this narrows the source-cue confound without settling it.

B Report-generation templates, verbatim

The complete text-generating logic of the deployed system, `app.py` lines 1645–1680 (function `predict_image`), verbatim. The four `descriptions` strings and the five heatmap branches are the entire vocabulary the interface can produce. No line reads a coordinate, area, side, or component count.

```
descriptions = {
    "Normal":       "No abnormal opacities detected in the lung fields.",
    "Pneumonia":    "Findings consistent with pneumonia. Warmer areas indicate possible consolidation.",
    "Tuberculosis": "Findings consistent with tuberculosis. Warmer areas indicate possible focal lesions
     or cavitation.",
    "Covid-19":     "Findings consistent with Covid-19. Warmer areas indicate possible bilateral
     ground-glass opacities.",
}

if is_uncertain:
    txt = (f"**Prediction:** Uncertain\n\n"
           f"The top class is {top_cls} at {top_prob * 100:.1f}%, below the "
           f"{CONFIDENCE_THRESHOLD * 100:.0f}% reporting threshold. Review the class "
           f"confidence values; a repeat or higher quality image may help.")
else:
    txt = (f"**Prediction:** {top_cls} ({top_prob * 100:.1f}% confidence)\n\n"
           f"{descriptions[top_cls]}")

# Explain what the heatmap represents for this specific case.
if mask is None:
    txt += "\n\nNo attention map is available for this model."
elif is_confident_finding:
    txt += (f"\n\nHeatmap: region driving the {viz_cls} prediction. "
            f"The outline marks the most influential area.")
elif is_auto and top_idx == 0:
    txt += (f"\n\nHeatmap: areas the model assessed for {viz_cls} "
            f"(the next most likely class); none reached an abnormal level.")
elif is_uncertain:
    txt += (f"\n\nHeatmap: areas the model weighed for {viz_cls}. "
            f"Interpret with caution while the prediction is uncertain.")
else:
    txt += f"\n\nHeatmap: model attention for {viz_cls} ({viz_prob * 100:.1f}%)."

if mask is not None:
    txt += ("\n\n_Overlay source: disease-segmentation U-Net._"
            if mask_source == "segmentation"
            else "\n\n_Overlay source: Grad-CAM._")
```

`CONFIDENCE_THRESHOLD = 0.60` (`app.py:67`); below it the first branch fires. The only other non-standard path is a caught exception that returns an empty string and a Gradio warning. `mask_source` is a two-word provenance label naming which network produced the overlay, never what it contains.

C Qualitative examples

Films from the 245-film pool, S0/S1 text verbatim from `tier1_summaries.csv`.

Phantom-outline claims (both films where `is_confident_finding` is true but no region survives the threshold; both Montgomery TB group).

- `194_MCUCXR_0015_0`: true Normal, predicted Tuberculosis at 95.7%. S0: "…The outline marks the most influential area. Overlay source: disease-segmentation U-Net." No region exists in the image. S1 under H_actual: "…No heatmap region met the reporting threshold; no location is reported." A confidently wrong call compounded by a claim about an outline that was never drawn.
- `202_MCUCXR_0301_1`: true and predicted Tuberculosis at 94.0%, classification correct, same phantom-outline pattern.

Well-localised. `209_Normal-2399`: true and predicted Normal at 100.0%, single focal region, upper-left zone, 93.6% inside the lung. S1: "…Heatmap evidence: left lung, upper zone, focal,

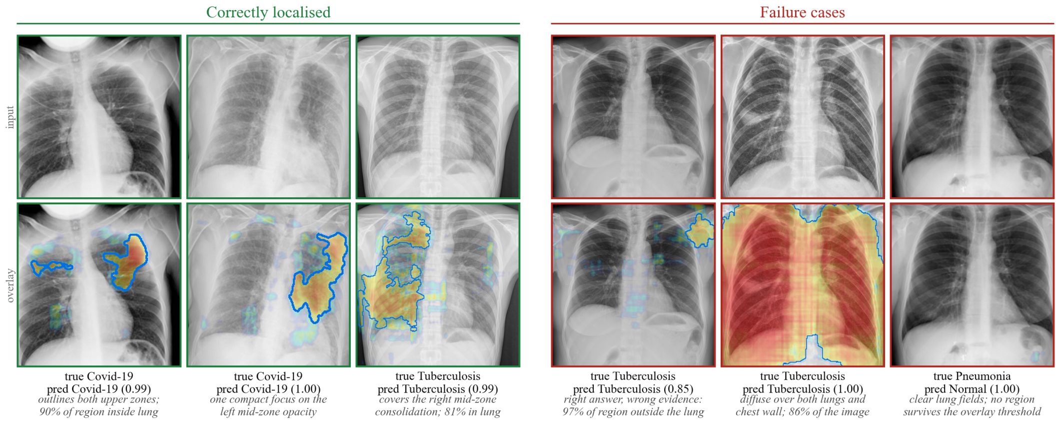

Figure 5: Six served overlays, input above overlay, spanning the range Appendix D quantifies. Left (green): compact regions that settle inside the lung field and follow the visible opacity, 66–90% of the region inside the lung. Right (red): a correct tuberculosis call whose evidence sits almost entirely in soft tissue at the apex (97% outside the lung); a region covering 86% of the image, spanning both lungs and the chest wall; and a clear, well-aerated film the model calls Normal, where no region survives the display threshold at all. All six overlays came from the served U-Net path. Per-case labels give the true class, the prediction, its confidence, and what the region did; italics are this appendix's reading, not part of the served text.

single, consistent with the normal finding."

Diffuse overlay. `011_person730_virus_1351`: true Pneumonia, predicted Normal at 96.0%. The region is extensive and multifocal, centroid outside both lungs, yet 50.3% of its area is inside the lung field, just over the gate. S1: "outside both, mid zone, extensive, multifocal," a literal description of a poorly localised region on a film the classifier got wrong, and a boundary case worth flagging.

No region. `003_person1175_virus_1981`: true Pneumonia, predicted Normal at 81.1%. No region survives the threshold. S0 gives no hint ("…none reached an abnormal level."); S1 states plainly that no location is reported.

D Additional qualitative statistics

Same 245-film pool. 41 (16.7%) produce no region above the overlay threshold; geometry percentages are over the remaining 204.

- *Activation leaving the lung.* 55% of overlays place the majority of the region outside the lung fields; 20% place almost all of it there, usually on the cardiac silhouette, the mediastinum, or below the diaphragm. The centroid is on neither lung in 129 of 204.
- *Edge activation.* 54% of regions reach the outer 2% of the frame despite the 8% crop; 14% put more than a quarter of the region in the outer tenth.
- *Diffuse activation.* 44% of regions cover more than a quarter of the image, 19% more than half, and 37% span both lung fields. This concentrates in Shenzhen (74% of its overlays diffuse; median TB-positive region 51% of the image); Montgomery fails oppositely (no region on a quarter of films; median region 4% where one exists).

On the 102 pneumonia-to-Normal errors, 28 produce no region and none produces a region over half the image; median region area is 11%, against 41% for correctly classified pneumonia. This is

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consistent with the label-noise reading: the lung fields are frequently clear and the small regions the model does produce plausibly agree with unremarkable pixels.

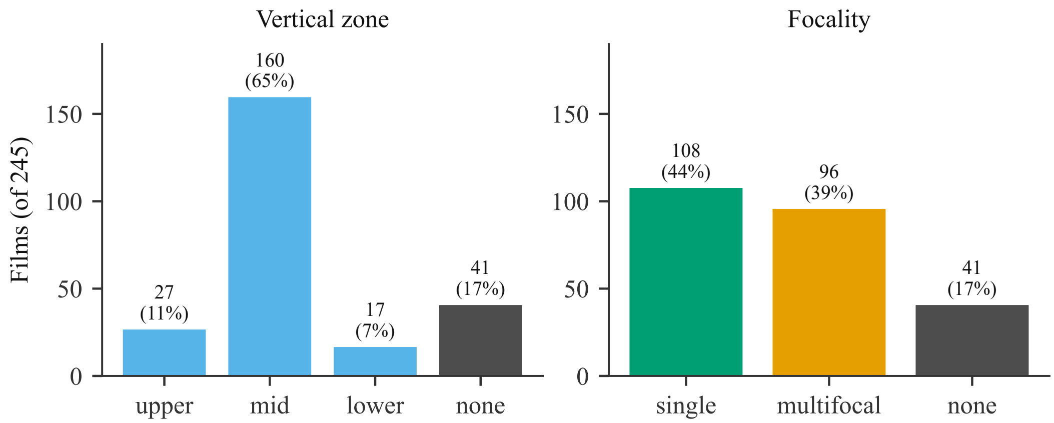

Figure 6: The two region descriptors added for this evaluation (Section 4), over all 245 films. Two thirds of regions centre in the mid zone; most of the rest produce no region at all, rather than an upper- or lower-zone one. Single- and multi-component regions are close to a coin flip, which is why focality is reported as its own descriptor rather than folded into extent.

E Latency, per stage

Table 4 splits the mean CPU wall-clock time reported in Section 5 by which localisation path served the request (i7-13620H, 10 PyTorch threads, single request). The two rows are two different code paths, not one path with variance: the served U-Net returns a mask in one forward pass; the live Grad-CAM fallback needs a forward pass and a backward pass through the classifier. Report generation is a dictionary lookup and does not appear as a separate row because it does not register above measurement noise.

Table 4: CPU latency by localisation path, i7-13620H, 10 threads, single request.

Path	Share of test films	Mean (sd), ms
Distilled U-Net (one forward pass)	91%	293 (81)
Live Grad-CAM fallback (forward + backward)	9%	560 (22)
Weighted (actual test-set mix)	100%	318

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NeurIPS paper checklist

1. **Claims.** Do the abstract and introduction state claims that match the paper's contributions and scope?
   [Yes] The central claim, that the deployed summary does not read the mask, is a source-level fact quoted verbatim in Section 3.2 and Appendix B. The abstract and Section 6 state which evaluation-grid cells are measured and which are proposed.

2. **Limitations.** Does the paper discuss the limitations of the work?
   [Yes] Section 6: unrun cells (H2, H4, S3/S4), no clinician ratings, dataset-source confounding, label noise in the dominant error mode, and the reproducibility fragility of the split.

3. **Theory.** Are assumptions and proofs given for every theoretical result?
   [NA] The paper reports measurements, not theorems. The one derived bound (localisation quality upper-bounds spatial-claim correctness) is stated with the Dice measurement it rests on.

4. **Experimental reproducibility.** Is there enough information to reproduce the main results?
   [Yes] The audited system is public (repository withheld for double blind). The descriptor extractor, the summary reconstructions, the thresholds, and the derangement seed are in Section 4; the 245-film scoring table and mask arrays are the system's own published analysis outputs.

5. **Open access to data and code.** Are data and code available?
   [Yes] The audited system and its analysis outputs are public. No new model was trained.

6. **Experimental settings.** Are training and test details specified?
   [Yes] For the components this paper adds there is no training (Section 4). The fixed upstream classifier and U-Net settings are in Section 3 and Appendix A.

7. **Error bars.** Are error bars or other measures of statistical significance reported?
   [Yes] Distributions (latency, Dice) carry a mean and standard deviation or a median. Single-number rates are exact counts over the full 245-film pool, so a confidence interval would describe sampling that did not occur.

8. **Compute.** Is the compute for the experiments reported?
   [Yes] The analysis is geometry over saved mask arrays and runs in seconds on a CPU; latency figures are from one laptop CPU (i7-13620H, 10 threads). No GPU, no training.

9. **Code of ethics.** Does the research conform to the NeurIPS Code of Ethics?
   [Yes] A retrospective audit of an existing research and educational prototype; no patient contact; the interface already carries a non-diagnostic disclaimer this paper argues should be more prominent.

10. **Broader impacts.** Are potential positive and negative societal impacts discussed?
    [Yes] Section 6 (Broader impact): the persuasiveness of a label plus a plausible heatmap plus a fluent summary, and why this paper is a caution against it.

11. **Safeguards.** Are safeguards described for any high-risk release?
    [NA] No models, datasets, or scrapers are released.

12. **Licenses.** Are existing assets credited and their licenses respected?
    [Yes] Backbone, localisation method, and segmentation architecture are cited in Section 3. The eight source datasets are public Kaggle datasets, named by their exact identifiers so each license can be consulted on its own page:

    - `tawsifurrahman/tuberculosis-tb-chest-xray-dataset`
    - `pcbreviglieri/pneumonia-xray-images`
    - `raddar/ricord-covid19-xray-positive-tests`

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- `tawsifurrahman/covid19-radiography-database`
- `raddar/tuberculosis-chest-xrays-shenzhen`
- `raddar/tuberculosis-chest-xrays-montgomery`
- `vbookshelf/tbx11k-simplified`
- `legrande/tbdata`

Terms differ per dataset; the camera-ready should transcribe each from its source page.

13. **New assets.** Are new assets documented?
    [NA] No new assets are released with this submission.

14. **Crowdsourcing and human subjects.** Are crowdsourcing and IRB details given?
    [NA] No human subjects, no crowdsourcing, no clinician annotation was collected.

15. **IRB / risk for human subjects.** Were risks disclosed and approvals obtained?
    [NA] No human-subjects research was conducted.

16. **LLM usage.** Is any LLM use in the core method declared?
    [NA] No LLM is used in the method. S2/S3/S4 LLM conditions are named as out of scope and not run.